PES University, Bangalore

Engineering Mathematics – I (UE26MA141A)

Unit 1: Partial Differentiation

Assignment — Step-by-step Solutions

Problems on partial derivatives

1. Find the first order partial derivatives of (i) $u = \tan^{-1}\dfrac{x^2+y^2}{x+y}$ (ii) $u = \cos^{-1}\left(\dfrac{x}{y}\right)$.

Solution (i)

Step 1. Put $w = \dfrac{x^2+y^2}{x+y}$, so $u = \tan^{-1} w$ and $\dfrac{\partial u}{\partial x} = \dfrac{1}{1+w^2}\dfrac{\partial w}{\partial x}$.

Step 2. By the quotient rule,

$$\frac{\partial w}{\partial x} = \frac{2x(x+y) - (x^2+y^2)\cdot 1}{(x+y)^2} = \frac{x^2+2xy-y^2}{(x+y)^2}.$$

Step 3. Also $1 + w^2 = \dfrac{(x+y)^2 + (x^2+y^2)^2}{(x+y)^2}$. Hence

$$\frac{\partial u}{\partial x} = \frac{(x+y)^2}{(x+y)^2+(x^2+y^2)^2}\cdot\frac{x^2+2xy-y^2}{(x+y)^2} = \frac{x^2+2xy-y^2}{(x+y)^2+(x^2+y^2)^2}.$$

Step 4. u is symmetric in x and y, so interchanging $x \leftrightarrow y$:

$$\frac{\partial u}{\partial y} = \frac{y^2+2xy-x^2}{(x+y)^2+(x^2+y^2)^2}.$$

Solution (ii)

Step 1. Recall $\dfrac{d}{dt}\cos^{-1} t = -\dfrac{1}{\sqrt{1-t^2}}$. With $t = x/y$ (take $y>0$): $\sqrt{1-\dfrac{x^2}{y^2}} = \dfrac{\sqrt{y^2-x^2}}{y}$.

Step 2.

$$\frac{\partial u}{\partial x} = -\frac{y}{\sqrt{y^2-x^2}}\cdot\frac{1}{y} = -\frac{1}{\sqrt{y^2-x^2}}.$$

Step 3.

$$\frac{\partial u}{\partial y} = -\frac{y}{\sqrt{y^2-x^2}}\cdot\left(-\frac{x}{y^2}\right) = \frac{x}{y\sqrt{y^2-x^2}}.$$

(i) $u_x = \dfrac{x^2+2xy-y^2}{(x+y)^2+(x^2+y^2)^2}$, $u_y = \dfrac{y^2+2xy-x^2}{(x+y)^2+(x^2+y^2)^2}$; (ii) $u_x = -\dfrac{1}{\sqrt{y^2-x^2}}$, $u_y = \dfrac{x}{y\sqrt{y^2-x^2}}$.

2. If $u = x^y$, show that $\dfrac{x}{y}\dfrac{\partial u}{\partial x} + \dfrac{1}{\log x}\dfrac{\partial u}{\partial y} = 2u$.

Solution

Step 1. Treating y as constant (power rule): $\dfrac{\partial u}{\partial x} = y\,x^{y-1}$.

Step 2. Treating x as constant (exponential rule): $\dfrac{\partial u}{\partial y} = x^y \log x$.

Step 3. Substitute:

$$\frac{x}{y}\cdot y\,x^{y-1}+\frac{1}{\log x}\cdot x^y\log x = x^y + x^y = 2x^y = 2u. \qquad \blacksquare$$

3. Verify that $\dfrac{\partial^2 u}{\partial x\,\partial y} = \dfrac{\partial^2 u}{\partial y\,\partial x}$ for $u = x^3 + y^3 + 3axy$.

Solution

Step 1. $\dfrac{\partial u}{\partial x} = 3x^2 + 3ay, \quad \dfrac{\partial u}{\partial y} = 3y^2 + 3ax.$

Step 2. $\dfrac{\partial^2 u}{\partial y\,\partial x} = \dfrac{\partial}{\partial y}(3x^2 + 3ay) = 3a, \quad \dfrac{\partial^2 u}{\partial x\,\partial y} = \dfrac{\partial}{\partial x}(3y^2 + 3ax) = 3a.$

Step 3. Both mixed derivatives equal $3a$, hence verified. $\blacksquare$

4. If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, show that
(i) $\left(\dfrac{\partial}{\partial x} + \dfrac{\partial}{\partial y} + \dfrac{\partial}{\partial z}\right)^2 u = -\dfrac{9}{(x+y+z)^2}$; (ii) $u_{xx} + u_{yy} + u_{zz} + 2u_{yz} + 2u_{zx} + 2u_{xy} = -\dfrac{9}{(x+y+z)^2}$.

Solution

Step 1. $u_x = \dfrac{3x^2 - 3yz}{x^3 + y^3 + z^3 - 3xyz}$, $u_y = \dfrac{3y^2 - 3zx}{x^3 + y^3 + z^3 - 3xyz}$, $u_z = \dfrac{3z^2 - 3xy}{x^3 + y^3 + z^3 - 3xyz}$.

Step 2. Add, and use the identity $x^3 + y^3 + z^3 - 3xyz = (x+y+z)(x^2 + y^2 + z^2 - xy - yz - zx)$:

$$u_x + u_y + u_z = \frac{3(x^2 + y^2 + z^2 - xy - yz - zx)}{(x+y+z)(x^2 + y^2 + z^2 - xy - yz - zx)} = \frac{3}{x+y+z}.$$

Step 3. Let $D = \dfrac{\partial}{\partial x} + \dfrac{\partial}{\partial y} + \dfrac{\partial}{\partial z}$. Then

$$D^2 u = D\left(\frac{3}{x+y+z}\right) = -\frac{3}{(x+y+z)^2} - \frac{3}{(x+y+z)^2} - \frac{3}{(x+y+z)^2} = -\frac{9}{(x+y+z)^2}.$$

This proves (i).

Step 4. Expanding the operator (mixed partials are equal since u is smooth):

$$D^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + 2\frac{\partial^2}{\partial y \partial z} + 2\frac{\partial^2}{\partial z \partial x} + 2\frac{\partial^2}{\partial x \partial y}.$$

So the left side of (ii) is exactly $D^2 u = -\dfrac{9}{(x+y+z)^2}$. $\blacksquare$

5. If $\theta = t^n e^{-r^2/(4t)}$, find n such that $\dfrac{1}{r^2}\dfrac{\partial}{\partial r}\left(r^2 \dfrac{\partial \theta}{\partial r}\right) = \dfrac{\partial \theta}{\partial t}$.

Solution

Step 1. $\dfrac{\partial \theta}{\partial r} = t^n e^{-r^2/4t}\left(-\dfrac{2r}{4t}\right) = -\dfrac{r}{2}\,t^{n-1}e^{-r^2/4t}$, so $r^2\dfrac{\partial \theta}{\partial r} = -\dfrac{1}{2}\,t^{n-1}r^3 e^{-r^2/4t}$.

Step 2. Differentiate w.r.t. r (product rule):

$$\frac{\partial}{\partial r}\left(r^2\theta_r\right) = -\frac{1}{2}t^{n-1}\left[3r^2 - \frac{r^4}{2t}\right]e^{-r^2/4t}.$$

Dividing by r^2: $\text{LHS} = t^{n-1}e^{-r^2/4t}\left(-\dfrac{3}{2} + \dfrac{r^2}{4t}\right)$.

Step 3. $\dfrac{\partial \theta}{\partial t} = nt^{n-1}e^{-r^2/4t} + t^n e^{-r^2/4t}\cdot\dfrac{r^2}{4t^2} = t^{n-1}e^{-r^2/4t}\left(n + \dfrac{r^2}{4t}\right).$

Step 4. Equate and cancel $t^{n-1}e^{-r^2/4t}$:

$$\frac{r^2}{4t} - \frac{3}{2} = n + \frac{r^2}{4t} \implies n = -\frac{3}{2}.$$

$$\boxed{n = -\frac{3}{2}}$$

6. If $\dfrac{x^2}{a^2+u} + \dfrac{y^2}{b^2+u} + \dfrac{z^2}{c^2+u} = 1$, prove $\left(\dfrac{\partial u}{\partial x}\right)^2 + \left(\dfrac{\partial u}{\partial y}\right)^2 + \left(\dfrac{\partial u}{\partial z}\right)^2 = 2\left(x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + z\dfrac{\partial u}{\partial z}\right)$.

Solution

Step 1. Here u is a function of x, y, z. Differentiate the relation partially w.r.t. x:

$$\frac{2x}{a^2+u} - \left[\frac{x^2}{(a^2+u)^2} + \frac{y^2}{(b^2+u)^2} + \frac{z^2}{(c^2+u)^2}\right]\frac{\partial u}{\partial x} = 0.$$

Step 2. Let $V = \dfrac{x^2}{(a^2+u)^2} + \dfrac{y^2}{(b^2+u)^2} + \dfrac{z^2}{(c^2+u)^2}$. Then

$$u_x = \frac{2x}{(a^2+u)V}, \quad u_y = \frac{2y}{(b^2+u)V}, \quad u_z = \frac{2z}{(c^2+u)V}.$$

Step 3. LHS:

$$u_x^2 + u_y^2 + u_z^2 = \frac{4}{V^2}\left[\frac{x^2}{(a^2+u)^2} + \frac{y^2}{(b^2+u)^2} + \frac{z^2}{(c^2+u)^2}\right] = \frac{4}{V^2}\cdot V = \frac{4}{V}.$$

Step 4. RHS:

$$2(xu_x + yu_y + zu_z) = \frac{4}{V}\left[\frac{x^2}{a^2+u} + \frac{y^2}{b^2+u} + \frac{z^2}{c^2+u}\right] = \frac{4}{V}\cdot 1 = \frac{4}{V}.$$

LHS = RHS. ■

7. If $x = r\cos\theta,\ y = r\sin\theta$, show that $\dfrac{\partial^2 r}{\partial x^2} + \dfrac{\partial^2 r}{\partial y^2} = \dfrac{1}{r}$.

Solution

Step 1. $r^2 = x^2 + y^2$, so $2r\,r_x = 2x \Rightarrow r_x = \dfrac{x}{r}$; similarly $r_y = \dfrac{y}{r}$.

Step 2. $r_{xx} = \dfrac{r\cdot 1 - x\,r_x}{r^2} = \dfrac{r - x^2/r}{r^2} = \dfrac{r^2 - x^2}{r^3} = \dfrac{y^2}{r^3}$. Similarly $r_{yy} = \dfrac{x^2}{r^3}$.

Step 3. $r_{xx} + r_{yy} = \dfrac{x^2+y^2}{r^3} = \dfrac{r^2}{r^3} = \dfrac{1}{r}$. ■

8. If $u = \dfrac{1}{\sqrt{x^2+y^2+z^2}}$, prove that $u_{xx} + u_{yy} + u_{zz} = 0$.

Solution

Step 1. Let $r^2 = x^2+y^2+z^2$, so $u = r^{-1}$ and $r_x = x/r$. Then $u_x = -r^{-2}\cdot\dfrac{x}{r} = -x\,r^{-3}$.

Step 2. $u_{xx} = -r^{-3} - x(-3r^{-4})\dfrac{x}{r} = -\dfrac{1}{r^3} + \dfrac{3x^2}{r^5}$. Similarly $u_{yy} = -\dfrac{1}{r^3} + \dfrac{3y^2}{r^5}$, $u_{zz} = -\dfrac{1}{r^3} + \dfrac{3z^2}{r^5}$.

Step 3. Adding: $-\dfrac{3}{r^3} + \dfrac{3(x^2+y^2+z^2)}{r^5} = -\dfrac{3}{r^3} + \dfrac{3r^2}{r^5} = 0$. ■

Total derivative, Implicit function, Chain rule

9. If $x^3 + y^3 - 3axy = 0$, find $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$.

Solution

Step 1. Let $f = x^3 + y^3 - 3axy$. Then $f_x = 3x^2 - 3ay$, $f_y = 3y^2 - 3ax$ and

$$\frac{dy}{dx} = -\frac{f_x}{f_y} = -\frac{3x^2 - 3ay}{3y^2 - 3ax} = \frac{ay - x^2}{y^2 - ax}.$$

Step 2. For the second derivative use

$$\frac{d^2y}{dx^2} = -\frac{f_{xx}f_y^2 - 2f_{xy}f_xf_y + f_{yy}f_x^2}{f_y^3}, \qquad f_{xx} = 6x,\ f_{xy} = -3a,\ f_{yy} = 6y.$$

Step 3. Write $p = x^2 - ay$, $q = y^2 - ax$ so $f_x = 3p$, $f_y = 3q$. Numerator $= 54xq^2 + 54apq + 54yp^2 = 54\,(xq^2 + apq + yp^2)$, denominator $= 27q^3$.

Step 4. Expand:

$$\begin{aligned} xq^2 &= xy^4 - 2ax^2y^2 + a^2x^3 \\ yp^2 &= x^4y - 2ax^2y^2 + a^2y^3 \\ apq &= ax^2y^2 - a^2x^3 - a^2y^3 + a^3xy \end{aligned}$$

Sum $= xy(x^3 + y^3 - 3axy) + a^3xy = 0 + a^3xy = a^3xy$ (using the given equation).

Step 5.

$$\frac{d^2y}{dx^2} = -\frac{54a^3xy}{27(y^2 - ax)^3} = -\frac{2a^3xy}{(y^2 - ax)^3} = \frac{2a^3xy}{(ax - y^2)^3}.$$

$$\boxed{\frac{dy}{dx} = \frac{ay - x^2}{y^2 - ax},\quad \frac{d^2y}{dx^2} = \frac{2a^3xy}{(ax - y^2)^3}}$$

10. Find $\dfrac{dy}{dx}$ when $(\cos x)^y = (\sin y)^x$.

Solution

Step 1. Take logarithms: $y \log\cos x = x \log\sin y$. Let $f(x, y) = y\log\cos x - x\log\sin y = 0$.

Step 2. $f_x = y \cdot \dfrac{-\sin x}{\cos x} - \log\sin y = -y\tan x - \log\sin y.$

$f_y = \log\cos x - x \cdot \dfrac{\cos y}{\sin y} = \log\cos x - x\cot y.$

Step 3.

$$\frac{dy}{dx} = -\frac{f_x}{f_y} = \frac{y\tan x + \log\sin y}{\log\cos x - x\cot y}.$$

11. If $u = x\log(xy)$ where $x^3 + y^3 + 3xy = 1$, find $\dfrac{du}{dx}$.

Solution

Step 1. Total derivative: $\dfrac{du}{dx} = \dfrac{\partial u}{\partial x} + \dfrac{\partial u}{\partial y}\dfrac{dy}{dx}$.

Step 2. $u = x\log x + x\log y$, so $\dfrac{\partial u}{\partial x} = \log(xy) + 1$, $\dfrac{\partial u}{\partial y} = \dfrac{x}{y}$.

Step 3. From $x^3 + y^3 + 3xy = 1$: $3x^2 + 3y + (3y^2 + 3x)\dfrac{dy}{dx} = 0 \Rightarrow \dfrac{dy}{dx} = -\dfrac{x^2 + y}{y^2 + x}$.

Step 4.

$$\frac{du}{dx} = 1 + \log(xy) - \frac{x}{y}\cdot\frac{x^2 + y}{y^2 + x}.$$

12. If $u = x^3 + y^3$ where $x = a\cos t,\ y = b\sin t$, find $\dfrac{du}{dt}$ and verify.

Solution

Step 1 (chain rule). $\dfrac{du}{dt} = \dfrac{\partial u}{\partial x}\dfrac{dx}{dt} + \dfrac{\partial u}{\partial y}\dfrac{dy}{dt} = 3x^2(-a\sin t) + 3y^2(b\cos t).$

Step 2. Substituting x, y: $\dfrac{du}{dt} = -3a^3\cos^2 t\sin t + 3b^3\sin^2 t\cos t.$

Step 3 (verification). Directly, $u = a^3\cos^3 t + b^3\sin^3 t$, so

$$\frac{du}{dt} = 3a^3\cos^2 t(-\sin t) + 3b^3\sin^2 t\cos t = -3a^3\cos^2 t\sin t + 3b^3\sin^2 t\cos t.$$

Both methods agree. ■

13. If $z = f(x,y)$, $x = e^u\cos v$, $y = e^u\sin v$, show that (i) $y\dfrac{\partial z}{\partial u} + x\dfrac{\partial z}{\partial v} = e^{2u}\dfrac{\partial z}{\partial y}$; (ii) $\left(\dfrac{\partial z}{\partial x}\right)^2 + \left(\dfrac{\partial z}{\partial y}\right)^2 = e^{-2u}\left[\left(\dfrac{\partial z}{\partial u}\right)^2 + \left(\dfrac{\partial z}{\partial v}\right)^2\right]$.

Solution

Step 1. $x_u = e^u\cos v = x$, $y_u = e^u\sin v = y$, $x_v = -e^u\sin v = -y$, $y_v = e^u\cos v = x$. Also $x^2 + y^2 = e^{2u}$.

Step 2 (chain rule).

$$z_u = z_x x_u + z_y y_u = x\,z_x + y\,z_y, \qquad z_v = z_x x_v + z_y y_v = -y\,z_x + x\,z_y.$$

Step 3 (i). $y z_u + x z_v = xy\,z_x + y^2 z_y - xy\,z_x + x^2 z_y = (x^2+y^2)z_y = e^{2u}z_y$. ■

Step 4 (ii).

$$z_u^2 + z_v^2 = (xz_x + yz_y)^2 + (-yz_x + xz_y)^2 = (x^2+y^2)(z_x^2 + z_y^2) = e^{2u}(z_x^2 + z_y^2)$$

(the cross terms $\pm 2xyz_xz_y$ cancel). Dividing by e^{2u} gives (ii). ■

14. If $u = x^2 + 2xy - y\ln z$, where $x = s + t^2$, $y = s - t^2$, $z = 2t^2$, find $\dfrac{\partial u}{\partial s}, \dfrac{\partial u}{\partial t}$ at $(1, 2, 1)$.

Solution

Step 1. $u_x = 2x + 2y$, $u_y = 2x - \ln z$, $u_z = -\dfrac{y}{z}$.

Step 2. $x_s = 1,\ y_s = 1,\ z_s = 0;\quad x_t = 2t,\ y_t = -2t,\ z_t = 4t.$

Step 3.

$$\frac{\partial u}{\partial s} = u_x\cdot 1 + u_y\cdot 1 + u_z\cdot 0 = 4x + 2y - \ln z.$$

$$\frac{\partial u}{\partial t} = (2x+2y)(2t) + (2x - \ln z)(-2t) + \left(-\frac{y}{z}\right)(4t) = 2t\left(2y + \ln z - \frac{2y}{z}\right).$$

Step 4. At $(x, y, z) = (1, 2, 1)$, $\ln 1 = 0$:

$$\frac{\partial u}{\partial s} = 4 + 4 - 0 = 8, \qquad \frac{\partial u}{\partial t} = 2t\,(4 + 0 - 4) = 0.$$

$$\boxed{\frac{\partial u}{\partial s} = 8,\ \ \frac{\partial u}{\partial t} = 0}$$

15. For the curve $xy^3 - yx^3 = 6$, find the slope and equation of the tangent at $(1, 2)$.

Solution

Step 1. Check: $1\cdot 8 - 2\cdot 1 = 6$, so $(1,2)$ lies on the curve. Let $f = xy^3 - yx^3 - 6$.

Step 2. $f_x = y^3 - 3x^2y$, $f_y = 3xy^2 - x^3$. At $(1,2)$: $f_x = 8 - 6 = 2$, $f_y = 12 - 1 = 11$.

Step 3. Slope $m = \dfrac{dy}{dx} = -\dfrac{f_x}{f_y} = -\dfrac{2}{11}$.

Step 4. Tangent: $y - 2 = -\dfrac{2}{11}(x-1) \Rightarrow 11y - 22 = -2x + 2 \Rightarrow 2x + 11y - 24 = 0$.

Slope $= -\dfrac{2}{11}$; tangent: $2x + 11y - 24 = 0$

16. If $w = f(x,y)$, $x = r\cos\theta$, $y = r\sin\theta$, show that $\left(\dfrac{\partial w}{\partial r}\right)^2 + \dfrac{1}{r^2}\left(\dfrac{\partial w}{\partial \theta}\right)^2 = \left(\dfrac{\partial w}{\partial x}\right)^2 + \left(\dfrac{\partial w}{\partial y}\right)^2$.

Solution

Step 1. $x_r = \cos\theta,\ y_r = \sin\theta,\ x_\theta = -r\sin\theta,\ y_\theta = r\cos\theta$.

Step 2.

$$w_r = w_x\cos\theta + w_y\sin\theta, \qquad \frac{1}{r}w_\theta = -w_x\sin\theta + w_y\cos\theta.$$

Step 3. Square and add; cross terms $\pm 2w_xw_y\sin\theta\cos\theta$ cancel:

$$w_r^2 + \frac{1}{r^2}w_\theta^2 = w_x^2(\cos^2\theta + \sin^2\theta) + w_y^2(\sin^2\theta + \cos^2\theta) = w_x^2 + w_y^2. \quad \blacksquare$$

Homogeneous Function and Euler's Theorem

Tools used. If f is homogeneous of degree n: $xf_x + yf_y = nf$ (Euler). If $f(u)$ is homogeneous of degree n, then $xu_x + yu_y = G(u) = n\dfrac{f(u)}{f'(u)}$, and $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = G(u)\,[G'(u) - 1]$.

Proof of the second formula: differentiate $xu_x + yu_y = G(u)$ w.r.t. x and multiply by x: $xu_x + x^2u_{xx} + xyu_{xy} = G'\,xu_x$. Likewise w.r.t. y, times y: $yu_y + xyu_{xy} + y^2u_{yy} = G'\,yu_y$. Adding: $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = (G'-1)(xu_x + yu_y) = G(G'-1)$.

17. Verify Euler's theorem for $z = \dfrac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}}$.

Solution

Step 1 (degree). $z(tx, ty) = \dfrac{t^{1/3}(x^{1/3} + y^{1/3})}{t^{1/2}(x^{1/2} + y^{1/2})} = t^{-1/6}z$, so z is homogeneous of degree $n = -\frac{1}{6}$. Euler predicts $xz_x + yz_y = -\frac{1}{6}z$.

Step 2. Let $N = x^{1/3} + y^{1/3}$, $D = x^{1/2} + y^{1/2}$. Then

$$z_x = \frac{\frac{1}{3}x^{-2/3}D - \frac{1}{2}x^{-1/2}N}{D^2} \ \Rightarrow\ xz_x = \frac{\frac{1}{3}x^{1/3}D - \frac{1}{2}x^{1/2}N}{D^2}, \quad yz_y = \frac{\frac{1}{3}y^{1/3}D - \frac{1}{2}y^{1/2}N}{D^2}.$$

Step 3. Adding:

$$xz_x + yz_y = \frac{\frac{1}{3}ND - \frac{1}{2}ND}{D^2} = -\frac{1}{6}\cdot\frac{N}{D} = -\frac{1}{6}z.$$

This agrees with Euler's theorem. $\blacksquare$

18. If $u = \cos^{-1}\left(\dfrac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, show that $xu_x + yu_y + \frac{1}{2}\cot u = 0$.

Solution

Step 1. $f = \cos u = \dfrac{x+y}{\sqrt{x}+\sqrt{y}}$ is homogeneous of degree $1 - \frac{1}{2} = \frac{1}{2}$.

Step 2. Euler: $xf_x + yf_y = \frac{1}{2}f$. Since $f_x = -\sin u\, u_x$, $f_y = -\sin u\, u_y$:

$$-\sin u\,(xu_x + yu_y) = \tfrac{1}{2}\cos u \;\Rightarrow\; xu_x + yu_y = -\tfrac{1}{2}\cot u.$$

Hence $xu_x + yu_y + \frac{1}{2}\cot u = 0$. ■

19. If $u = \sin^{-1}\left[\dfrac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}\right]$, show that $xu_x + yu_y + zu_z + 3\tan u = 0$.

Solution

Step 1. $f = \sin u = \dfrac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$; numerator has degree 1, denominator degree 4, so f is homogeneous of degree -3.

Step 2. Euler: $xf_x + yf_y + zf_z = -3f$, i.e. $\cos u\,(xu_x + yu_y + zu_z) = -3\sin u$.

Step 3. So $xu_x + yu_y + zu_z = -3\tan u$, i.e. $xu_x + yu_y + zu_z + 3\tan u = 0$. ■

20. If $u = \tan^{-1}\left(\dfrac{x^3+y^3}{x-y}\right)$, prove (i) $xu_x + yu_y = \sin 2u$; (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = 2\cos 3u \sin u$.

Solution

Step 1. $f = \tan u = \dfrac{x^3+y^3}{x-y}$ is homogeneous of degree $3-1 = 2$.

Step 2 (i). Euler: $\sec^2 u\,(xu_x + yu_y) = 2\tan u \Rightarrow xu_x + yu_y = 2\dfrac{\sin u}{\cos u}\cos^2 u = 2\sin u\cos u = \sin 2u$. ■

Step 3 (ii). With $G(u) = \sin 2u$, $G'(u) = 2\cos 2u$:

$$x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = G(G'-1) = \sin 2u\,(2\cos 2u - 1) = \sin 4u - \sin 2u.$$

Step 4. Using $\sin A - \sin B = 2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$: $\sin 4u - \sin 2u = 2\cos 3u \sin u$. ■

21. If $u = \sin^{-1}\left(\dfrac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, prove $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = -\dfrac{\sin u\cos 2u}{4\cos^3 u}$.

Solution

Step 1. $f = \sin u = \dfrac{x+y}{\sqrt{x}+\sqrt{y}}$ is homogeneous of degree $\frac{1}{2}$.

Step 2. Euler: $\cos u\,(xu_x + yu_y) = \frac{1}{2}\sin u \Rightarrow G(u) = xu_x + yu_y = \frac{1}{2}\tan u$.

Step 3. $G'(u) = \frac{1}{2}\sec^2 u$, so $G' - 1 = \dfrac{1 - 2\cos^2 u}{2\cos^2 u} = -\dfrac{\cos 2u}{2\cos^2 u}$.

Step 4.

$$x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = G(G'-1) = \frac{1}{2}\cdot\frac{\sin u}{\cos u}\cdot\left(-\frac{\cos 2u}{2\cos^2 u}\right) = -\frac{\sin u\cos 2u}{4\cos^3 u}. \quad ■$$

22. If $u = \sin^{-1}\sqrt{x^2+y^2}$, show (i) $xu_x + yu_y = \tan u$; (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = \tan^3 u$.

Solution

Step 1. $f = \sin u = \sqrt{x^2+y^2}$ is homogeneous of degree 1.

Step 2 (i). Euler: $\cos u\,(xu_x + yu_y) = \sin u \Rightarrow xu_x + yu_y = \tan u$. ■

Step 3 (ii). $G = \tan u$, $G' = \sec^2 u$:

$$x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = \tan u(\sec^2 u - 1) = \tan u\cdot\tan^2 u = \tan^3 u. \quad ■$$

Taylor and Maclaurin Series of $f(x, y)$

Formula. With $h = x - a$, $k = y - b$ and all derivatives evaluated at (a, b):

$$f(x,y) = f + (hf_x + kf_y) + \frac{1}{2!}(h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}) + \frac{1}{3!}(h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}) + \cdots$$

For Maclaurin series, $(a, b) = (0, 0)$.

23. Expand $e^x \sin y$ in ascending powers of x and y.

Solution

Step 1. Derivatives and their values at $(0, 0)$:

$$\begin{array}{lll} f = e^x \sin y = 0 & f_x = e^x \sin y = 0 & f_y = e^x \cos y = 1 \\ f_{xx} = e^x \sin y = 0 & f_{xy} = e^x \cos y = 1 & f_{yy} = -e^x \sin y = 0 \\ f_{xxx} = 0 & f_{xxy} = e^x \cos y = 1 & f_{xyy} = -e^x \sin y = 0, \quad f_{yyy} = -e^x \cos y = -1 \end{array}$$

Step 2. Substitute:

$$e^x \sin y = 0 + [x \cdot 0 + y \cdot 1] + \frac{1}{2!}[x^2 \cdot 0 + 2xy \cdot 1 + y^2 \cdot 0] + \frac{1}{3!}[x^3 \cdot 0 + 3x^2 y \cdot 1 + 3xy^2 \cdot 0 + y^3(-1)] + \cdots$$

$$\boxed{e^x \sin y = y + xy + \frac{1}{2}x^2 y - \frac{1}{6}y^3 + \cdots}$$

24. Expand $\tan^{-1}\dfrac{y}{x}$ about $(1, 1)$ up to second degree terms; hence compute $f(1.1, 0.9)$.

Solution

Step 1. Let $\rho = x^2 + y^2$ ($\rho = 2$ at $(1,1)$). $f(1,1) = \tan^{-1} 1 = \dfrac{\pi}{4}$.

Step 2 (first order). $f_x = \dfrac{-y}{\rho} = -\dfrac{1}{2}$, $\quad f_y = \dfrac{x}{\rho} = \dfrac{1}{2}$.

Step 3 (second order). $f_{xx} = \dfrac{2xy}{\rho^2} = \dfrac{1}{2}$, $\quad f_{xy} = \dfrac{y^2 - x^2}{\rho^2} = 0$, $\quad f_{yy} = \dfrac{-2xy}{\rho^2} = -\dfrac{1}{2}$.

Step 4 (third order, for better accuracy). $f_{xxx} = \dfrac{2y}{\rho^2} - \dfrac{8x^2 y}{\rho^3} = -\dfrac{1}{2}$, $f_{xxy} = \dfrac{2x}{\rho^2} - \dfrac{8xy^2}{\rho^3} = -\dfrac{1}{2}$, $f_{xyy} = -\dfrac{2y}{\rho^2} + \dfrac{8x^2 y}{\rho^3} = \dfrac{1}{2}$, $f_{yyy} = -\dfrac{2x}{\rho^2} + \dfrac{8xy^2}{\rho^3} = \dfrac{1}{2}$.

Step 5. With $h = x - 1,\ k = y - 1$:

$$\tan^{-1}\frac{y}{x} = \frac{\pi}{4} - \frac{1}{2}h + \frac{1}{2}k + \frac{1}{4}h^2 - \frac{1}{4}k^2 - \frac{1}{12}\left[h^3 + 3h^2 k - 3hk^2 - k^3\right] + \cdots$$

Step 6. At $(1.1, 0.9)$: $h = 0.1$, $k = -0.1$.
First order: $-\frac{1}{2}(0.1) + \frac{1}{2}(-0.1) = -0.1$. Second order: $\frac{1}{4}(0.01) - \frac{1}{4}(0.01) = 0$.
Third order: $h^3 + 3h^2 k - 3hk^2 - k^3 = 0.001 - 0.003 - 0.003 + 0.001 = -0.004$, giving $-\frac{1}{12}(-0.004) = +0.000333$.

Step 7. Up to second degree: $f \approx 0.785398 - 0.1 = 0.6854$. Including third-degree terms: $f \approx 0.6854 + 0.0003 = 0.6857$.

$$\boxed{f(1.1, 0.9) \approx 0.6857 \quad \text{(exact value } \tan^{-1}(9/11) = 0.68573\ldots\text{)}}$$

The printed answer key shows the $(x-1)^2$ coefficient as $-\frac{1}{2}$; the correct value is $f_{xx}(1,1) = +\frac{1}{2}$, which is also what the key uses in its numerical step ($+\frac{1}{4}(0.1)^2$).

25. Expand $x^2 y + 3y - 2$ in powers of $(x - 1)$ and $(y + 2)$.

Solution

Step 1. Expand about $(a, b) = (1, -2)$; $h = x - 1$, $k = y + 2$. $f(1, -2) = -2 - 6 - 2 = -10$.

Step 2. $f_x = 2xy = -4$, $f_y = x^2 + 3 = 4$; $f_{xx} = 2y = -4$, $f_{xy} = 2x = 2$, $f_{yy} = 0$; $f_{xxy} = 2$, all other third (and higher) derivatives are 0.

Step 3.

$$f = -10 + (-4h + 4k) + \frac{1}{2!}(-4h^2 + 2 \cdot 2hk + 0) + \frac{1}{3!}(3h^2 k \cdot 2)$$

$$x^2y + 3y - 2 = -10 - 4(x-1) + 4(y+2) - 2(x-1)^2 + 2(x-1)(y+2) + (x-1)^2(y+2)$$

26. Expand $f(x, y) = e^x \log(1 + y)$ in powers of x and y up to third degree.

Solution

Step 1. Values at $(0, 0)$:

$$f = 0 \qquad f_x = e^x \log(1+y) = 0 \qquad f_y = \frac{e^x}{1+y} = 1$$
$$f_{xx} = 0 \qquad f_{xy} = \frac{e^x}{1+y} = 1 \qquad f_{yy} = -\frac{e^x}{(1+y)^2} = -1$$
$$f_{xxx} = 0 \qquad f_{xxy} = 1,\ f_{xyy} = -1 \qquad f_{yyy} = \frac{2e^x}{(1+y)^3} = 2$$

Step 2.

$$f = y + \frac{1}{2!}(2xy - y^2) + \frac{1}{3!}(3x^2y - 3xy^2 + 2y^3) + \cdots$$

$$e^x \log(1+y) = y + xy - \frac{1}{2}y^2 + \frac{1}{2}(x^2y - xy^2) + \frac{1}{3}y^3 + \cdots$$

Check: $(1 + x + \frac{x^2}{2})(y - \frac{y^2}{2} + \frac{y^3}{3})$ gives the same terms. The y^3 coefficient is $+\frac{1}{3}$ (the printed key's $-\frac{1}{3}$ is a typo).

Maxima and Minima of $f(x, y)$

Test. At a stationary point ($f_x = f_y = 0$) let $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$. If $rt - s^2 > 0$: maximum when $r < 0$, minimum when $r > 0$. If $rt - s^2 < 0$: saddle point. If $rt - s^2 = 0$: further investigation needed.

27. Test $f(x, y) = x^3y^2(6 - x - y)$ for maxima and minima (points not at the origin).

Solution

Step 1. $f = 6x^3y^2 - x^4y^2 - x^3y^3$.

$f_x = 18x^2y^2 - 4x^3y^2 - 3x^2y^3 = x^2y^2(18 - 4x - 3y)$, $\qquad f_y = 12x^3y - 2x^4y - 3x^3y^2 = x^3y(12 - 2x - 3y)$.

Step 2. For $x \neq 0,\ y \neq 0$: $4x + 3y = 18$ and $2x + 3y = 12$. Subtracting, $2x = 6 \Rightarrow x = 3$, $y = 2$. Stationary point: $(3, 2)$.

Step 3.

$$r = f_{xx} = 36xy^2 - 12x^2y^2 - 6xy^3 = 432 - 432 - 144 = -144$$
$$s = f_{xy} = 36x^2y - 8x^3y - 9x^2y^2 = 648 - 432 - 324 = -108$$
$$t = f_{yy} = 12x^3 - 2x^4 - 6x^3y = 324 - 162 - 324 = -162$$

Step 4. $rt - s^2 = (-144)(-162) - (-108)^2 = 23328 - 11664 = 11664 > 0$ and $r < 0 \Rightarrow$ maximum.

Maximum at $(3, 2)$; maximum value $f(3, 2) = 27 \cdot 4 \cdot 1 = 108$.

28. Examine $f(x, y) = \sin x + \sin y + \sin(x + y)$ for maxima and minima.

Solution

Step 1. $f_x = \cos x + \cos(x+y) = 0$, $f_y = \cos y + \cos(x+y) = 0$. Subtracting: $\cos x = \cos y$, so (taking $0 < x, y < \pi$ as usual) $x = y$.

Step 2. Then $\cos x + \cos 2x = 0 \Rightarrow 2\cos^2 x + \cos x - 1 = 0 \Rightarrow (2\cos x - 1)(\cos x + 1) = 0$, so $\cos x = \frac{1}{2}$ or $\cos x = -1$, i.e. $x = \frac{\pi}{3}$ or $x = \pi$. Stationary points: $\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$, (π, π).

Step 3. $r = -\sin x - \sin(x+y)$, $s = -\sin(x+y)$, $t = -\sin y - \sin(x+y)$.

Step 4. At $\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$: $r = -\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} = -\sqrt{3}$, $s = -\frac{\sqrt{3}}{2}$, $t = -\sqrt{3}$. $rt - s^2 = 3 - \frac{3}{4} = \frac{9}{4} > 0$, $r < 0 \Rightarrow$ maximum.

Step 5. At (π, π): $r = s = t = 0$, so the test is inconclusive. Writing $x = \pi + h,\ y = \pi + k$: $f = -\sin h - \sin k + \sin(h+k) \approx -\frac{1}{2}hk(h+k)$, which changes sign near $(0,0)$, so (π, π) is neither a maximum nor a minimum.

Step 6.

$$f_{\max} = f\left(\tfrac{\pi}{3}, \tfrac{\pi}{3}\right) = \sin\tfrac{\pi}{3} + \sin\tfrac{\pi}{3} + \sin\tfrac{2\pi}{3} = \tfrac{\sqrt{3}}{2} + \tfrac{\sqrt{3}}{2} + \tfrac{\sqrt{3}}{2} = \tfrac{3\sqrt{3}}{2}.$$

Maximum at $\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$, maximum value $\dfrac{3\sqrt{3}}{2}$.

Lagrange's Method of Undetermined Multipliers

29. Find the maximum and minimum distances of the point $(3, 4, 12)$ from the sphere $x^2 + y^2 + z^2 = 1$.

Solution

Step 1. Extremise the squared distance $D = (x-3)^2 + (y-4)^2 + (z-12)^2$ subject to $\phi = x^2 + y^2 + z^2 - 1 = 0$. Let $F = D + \lambda\phi$.

Step 2.

$$F_x = 2(x-3) + 2\lambda x = 0 \Rightarrow x = \frac{3}{1+\lambda}, \quad y = \frac{4}{1+\lambda}, \quad z = \frac{12}{1+\lambda}.$$

Step 3. Substitute into the constraint: $\dfrac{9 + 16 + 144}{(1+\lambda)^2} = 1 \Rightarrow (1+\lambda)^2 = 169 \Rightarrow 1 + \lambda = \pm 13$.

Step 4. $1 + \lambda = 13$: $P\left(\frac{3}{13}, \frac{4}{13}, \frac{12}{13}\right)$. $1 + \lambda = -13$: $Q\left(-\frac{3}{13}, -\frac{4}{13}, -\frac{12}{13}\right)$.

Step 5. Distances from $A(3, 4, 12)$:

$$AP = \sqrt{\left(\tfrac{36}{13}\right)^2 + \left(\tfrac{48}{13}\right)^2 + \left(\tfrac{144}{13}\right)^2} = \frac{12}{13}\sqrt{169} = 12, \qquad AQ = \frac{14}{13}\sqrt{169} = 14.$$

(Check: A is at distance 13 from the centre, so 13 ∓ 1.)

Minimum distance 12 at $P\left(\frac{3}{13}, \frac{4}{13}, \frac{12}{13}\right)$; maximum distance 14 at $Q\left(-\frac{3}{13}, -\frac{4}{13}, -\frac{12}{13}\right)$.

30. Show that the rectangular solid of maximum volume that can be inscribed in a given sphere is a cube.

Solution

Step 1. Let the sphere be $x^2 + y^2 + z^2 = a^2$. An inscribed box centred at the origin has vertices $(\pm x, \pm y, \pm z)$, edges $2x, 2y, 2z$ and volume $V = 8xyz$ $(x, y, z > 0)$.

Step 2. $F = 8xyz + \lambda(x^2 + y^2 + z^2 - a^2)$:

$$F_x = 8yz + 2\lambda x = 0, \quad F_y = 8xz + 2\lambda y = 0, \quad F_z = 8xy + 2\lambda z = 0.$$

Step 3. Multiply by x, y, z respectively: $8xyz = -2\lambda x^2 = -2\lambda y^2 = -2\lambda z^2$. Since $xyz \neq 0$, $\lambda \neq 0$, so $x^2 = y^2 = z^2 \Rightarrow x = y = z$.

Step 4. From the constraint, $3x^2 = a^2 \Rightarrow x = y = z = \dfrac{a}{\sqrt{3}}$. All edges equal $\dfrac{2a}{\sqrt{3}}$ — the box is a **cube**.

Step 5 (why a maximum). $V \geq 0$, $V = 0$ when any edge is 0, and V is continuous on the closed bounded constraint set, so it attains a positive maximum at an interior critical point; the only one is $x = y = z$. Maximum volume $V = 8\left(\dfrac{a}{\sqrt{3}}\right)^3 = \dfrac{8a^3}{3\sqrt{3}}$. $\blacksquare$